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Polynomials



Polynomials have many practical uses. For example, they can be used to solve construction and material planning challenges. In any 2-D construction situation, a polynomial equation can be utilised to calculate the amount of materials required.



What's Inside

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TOPIC 1

INTRODUCTION TO POLYNOMIALS

Learning Objectives

- Students will be learning to identify the coefficients of polynomial, terms of polynomial and zeroes of polynomial.
- Students will be learning to explore the types of polynomials on the basis of terms and their degrees.

Learning Outcomes

- Students will be able to find the degree of polynomial by observing the highest power of variable.
- Students will be able to identify the coefficient of a polynomial term.

Real Life Application

We've all seen a roller coaster, whether on television or in real life. It is a type of amusement ride that features elevated tracks with tight twists, steep slopes, and inversions.



Have you ever wondered how engineers create such tracks while maintaining proper safety and enjoyment? They use the polynomial concept, in which the form of the tracks represents a polynomial equation, say in x , and thrill of the ride can be evaluated by inserting different values of x and computing the value of the polynomial.

Let us now gain more knowledge about polynomials.

Polynomials are derived from variables and constants, using operations such as addition, subtraction, multiplication, and taking non-negative integer powers. It can be expressed as the sum of a finite number of terms.

Polynomials in One Variable

Polynomial $p(x)$ in one variable x is an algebraic expression (in x) of the form $p(x) = a_n x^n + a_{(n-1)} x^{(n-1)} + a_{(n-2)} x^{(n-2)} + a_2 x^2 + a_1 x + a_0$.

Where $a_0, a_1, a_2, \dots, a_n$ are constants and $a_n \neq 0$.

Here $a_0, a_1, a_2, \dots, a_n$ are respectively the coefficients of $x^0, x^1, x^2, \dots, x^n$ and n is called the degree of the polynomial. Each of $a_n x^n, a_{n-1} x^{n-1}, \dots, a_0$, with $a_n \neq 0$ is called a term $p(x)$ of the polynomial.

This form of the polynomial is known as the “standard form of the polynomial”.

E.g., $3x^3 - 2x^2 + 7x - 3$ is a polynomial in one variable x .



Caution

→ The degree of a polynomial can only be the whole number, e.g., $3\sqrt{t} + t\sqrt{2}$ is not a polynomial as the power of t is $\frac{1}{2}$.

Example 1. Which of the following expressions are polynomials in one variable and which are not? State reasons for your answer.

(A) $4x^2 - 3x + 7$

(B) $y + \frac{2}{y}$

[NCERT]

Ans. (A) $4x^2 - 3x + 7 = 4x^2 - 3x^1 + 7x^0$

Here, powers of the equation are 2, 1 and 0.

Since, all powers are whole numbers, it is a polynomial.

Now, since there is only one variable x , it is polynomial in one variable.

$$(B) \quad y + \frac{2}{y} = y^1 + \frac{2}{y^1}$$

$$= y^1 + 2y^{-1} \quad \left[\text{As } \frac{1}{a^m} = a^{-m} \right]$$



Here, the powers of y are 1 and -1 .

Since, -1 is not a whole number, it is not a polynomial.

Coefficient of the Polynomial

The coefficient of the polynomial is a number placed in multiplication with the variable. Each term of a polynomial has a coefficient.

Example 2. Write the coefficient of x^2 in each of the following:

(A) $2 + x^2 + x$

(B) $2 - x^2 + x^3$

(C) $\frac{\pi}{2} x^2 + x$

[NCERT]

Ans. (A) $2 + x^2 + x$

$$\begin{aligned} &= 2 + x^2 + x \\ &= 2 + 1 \times x^2 + x \end{aligned}$$

↑ coefficient of x^2

∴ Coefficient of $x^2 = 1$

(B) $2 - x^2 + x^3$

$$= 2 + (-1) \times x^2 + x^3$$

↑ coefficient of x^2

∴ Coefficient of $x^2 = -1$

(C) $\frac{\pi}{2} x^2 + x$

$$= \frac{\pi}{2} \times x^2 + x$$

↑ coefficient of x^2

∴ Coefficient of $x^2 = \frac{\pi}{2}$

Constant Polynomial

A polynomial of degree zero, i.e., polynomial with only constant term is called constant polynomials.

E.g., $\frac{1}{4}$, 2, 3, 7 are constant polynomials.

Zero Polynomial

Zero polynomials are the polynomials whose coefficients are all equal to zero. It is the value of the variable that makes the polynomial equal to zero. The degree of the zero polynomial is not defined.

E.g., $0x^2 + 0x + 0$.

Degree of a Polynomial

The highest power of the variables in a polynomial is called the degree of the polynomial.

- (1) In a polynomial with one variable, the highest power of the variable is called the degree of the polynomial.

E.g., $7x^8 - 3x^5 + 2x + 1$ is a polynomial in x of degree 8.

- (2) In a polynomial with more than one variable, the sum of the powers of the variables in each term is taken up and the highest sum obtained is the degree of the polynomial.

E.g., $17x^8 - 5x^6y^4 + 9x^2y - 4y + 3$ is a polynomial in x and y of degree 10.

Terms

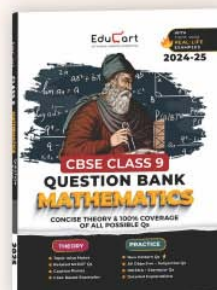
In a polynomial, the number of expressions are called terms.

E.g., $17x^3 - 3x^2 + 2x + 1$ is polynomial with four terms, i.e., $17x^3$, $-3x^2$, $2x$ and 1.



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OBJECTIVE Type Questions

Multiple Choice Questions

[1 mark each]

1. If $x^2 + kx + 6 = (x + 2)(x + 3)$ for all x , then the value of k is:

- (a) 1 (b) -1
(c) 5 (d) 3

[NCERT Exemplar]

Ans. (c) 5

Explanation:

Given equation is $x^2 + kx + 6 = (x + 2)(x + 3)$

On simplifying,

$$x^2 + kx + 6 = (x + 2)(x + 3)$$

$$x^2 + kx + 6 = x^2 + 3x + 2x + 6$$

$$x^2 + kx + 6 = x^2 + 5x + 6$$

On comparing, we get

$$kx = 5x$$

$$k = 5$$

Therefore, the value of k is 5.

2. If $x + y = 14$ and $xy = 24$, then the value of $x^2 - xy + y^2$ is:

- (a) 196 (b) 124
(c) -142 (d) 72

Ans. (b) 124

Explanation:

We know,

$$x^3 + y^3 = (x + y)(x^2 - xy + y^2) \quad \dots(i)$$

And $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$

We can write,

$$x^3 + y^3 = (x + y)^3 - 3xy(x + y) \quad \dots(ii)$$

Now, equating equations (i) and (ii)

$$\begin{aligned} (x + y)(x^2 - xy + y^2) \\ = (x + y)^3 - 3xy(x + y) \end{aligned}$$

$$\begin{aligned}
 (x^2 - xy + y^2) &= \frac{(x+y)^3 - 3xy(x+y)}{(x+y)} \\
 &= \frac{(x+y)[(x+y)^2 - 3xy]}{(x+y)} \\
 &= (x+y)^2 - 3xy \\
 &= (14)^2 - 3(24) \\
 &= 196 - 72 \\
 &= 124
 \end{aligned}$$

Assertion-Reason

[A-R]

[1 mark each]

In this following question, a statement of Assertion (A) is followed by a statement of Reason (R).

Choose the correct option as:

- (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).
- (b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).
- (c) Assertion (A) is true but reason (R) is false.
- (d) Assertion (A) is false but reason (R) is true.

3. Assertion (A): The degree of the polynomial $6a^2 - \frac{1}{8}a^6 + 34a^3 - \sqrt{12}a$ is 6.

Reason (R): The degree of a polynomial is the highest power of its variable.

Ans. (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

Explanation:

Given: $6a^2 - \frac{1}{8}a^6 + 34a^3 - \sqrt{12}a$

Here, the degree of this polynomial is 6 because the degree of a polynomial is the highest power of its variable.



CASE BASED Questions (CBQs)

[5 marks each]

Read the following passages and answer the questions that follow:

4. An object which is thrown or projected into the air, subject to only the acceleration of gravity is called a projectile and its path is called its trajectory. The curved path a projectile follows was shown by Galileo to be a parabola. Parabola is represented by a polynomial. If the polynomial to represent the distance covered is: $p(x) = -6x^2 + 48x + 24$.



(A) What is the degree of the polynomial?

- (a) 0 (b) 1
(c) 2 (d) 3

(B) Find the height of the projectile 5 seconds after it's launched.

- (a) 114 m (b) 64 m
(c) 170 m (d) 116 m

(C) The value of K if $x - 3$ is a factor of $K^2x^3 - x^2 + 3x - 1$, is:

- (a) 0 (b) $\pm \frac{1}{3\sqrt{3}}$
(c) $\frac{1}{\sqrt{3}}$ (d) $\frac{1}{\sqrt{2}}$

(D) If the polynomial to represent the distance covered is given by,
 $p(x) = -6x^2 + 48x + 24$, then the value of $p(4)$ is:

- (a) 117 (b) -72
 (c) 120 (d) 114

(E) If the equation of the parabola is given by, $p(y) = 3y^2 - 2y - 4$, then the value of $p(2)$ is:

- (a) 8 (b) 4
 (c) 12 (d) 6

Ans. (A) (c) 2

Explanation: The degree of the polynomial is 2.

(B) (a) 114 m

Explanation: $p(y) = -6y^2 + 48y + 24$

To calculate height,

$$\text{put } y = 5$$

Height of projectile,

$$\begin{aligned} p(5) &= -6 \times (5)^2 + 48 \times 5 + 24 \\ &= -6 \times 25 + 240 + 24 \\ &= 114 \end{aligned}$$

So, the height of projectile reaches is 114 m.

(C) (b) $\pm \frac{1}{3\sqrt{3}}$

Explanation: Let $p(x) = K^2x^3 - x^2 + 3x - 1$.

If $(x - 3)$ is a factor of $p(x)$ then $p(x) = 0$.

$$\begin{aligned} \therefore p(3) &= K^2(3)^3 - (3)^2 + 3(3) - 1 \\ &= K^2(27) - 9 + 9 - 1 \\ &= 27K^2 - 1 \end{aligned}$$

$$K^2 = \frac{1}{27}$$

$$K^2 = \pm \frac{1}{3\sqrt{3}}$$



(D) (c) 120

Explanation: $p(x) = 6x^2 + 48x + 24$

Put $x = 4$

$$\begin{aligned}\therefore p(4) &= 6(4)^2 + 48(4) + 24 \\ &= -96 + 192 + 24 \\ &= 120\end{aligned}$$

Hence, the value of $p(4)$ is 120.

(E) (b) 4

Explanation: $p(y) = 3y^2 - 2y - 4$

$$\begin{aligned}p(2) &= 3 \times (2)^2 - 2 \times 2 - 4 \\ &= 3 \times 4 - 4 - 4 \\ &= 12 - 8\end{aligned}$$

$$p(2) = 4$$

VERY SHORT ANSWER Type Questions (VSA)

[1 mark each]

5. If $p(x) = 2x^3 - 6x^2 + ax + a$, and $(x + 2)$ is a factor of $p(x)$, then find a .

Ans.

$$p(x) = 2x^3 - 6x^2 + ax + a$$

If $(x + 2)$ is a factor, $p(-2) = 0$

$$p(-2) = 0$$

$$2(-2)^3 - 6(-2)^2 + a(-2) + a = 0$$

$$\Rightarrow 2(-8) - 6(4) - 2a + a = 0$$

$$\Rightarrow -16 - 24 - a = 0$$

$$a = -40$$

SHORT ANSWER Type-I Questions (SA-I)

[2 marks each]

6. Ashima donated a certain amount of money to a blind school. Her friend Manya wanted to know the amount donated by her, but Ashima did not disclose the amount she donated, instead

she gave her a hint that if $\left(x + \frac{1}{x}\right) = ₹7$ then the amount donated by her is ₹ $\left(x^3 + \frac{1}{x^3}\right)$. Find the amount donated by Ashima to the school. [British Council 2022]

$$\begin{aligned} \text{Ans. } \therefore \quad & x + \frac{1}{x} = 7 \\ \therefore \quad & \left[x + \frac{1}{x}\right]^3 = 7^3 \\ \Rightarrow \quad & x^3 + \frac{1}{x^3} + 3x \cdot \frac{1}{x} \left(x + \frac{1}{x}\right) = 343 \\ \Rightarrow \quad & x^3 + \frac{1}{x^3} + 3 \times 1 \times (7) = 343 \\ \Rightarrow \quad & x^3 + \frac{1}{x^3} + 21 = 343 \\ \Rightarrow \quad & x^3 + \frac{1}{x^3} = 343 - 21 \\ & = 322 \end{aligned}$$

Thus, the amount donated by Ashima is ₹ 322.

SHORT ANSWER Type-II Questions (SA-II)

[3 marks each]

7. (A) Find the value of $48^3 - 30^3 - 18^3$ using suitable algebraic identities.
 (B) Factorise $(x - y)^3 + (y - z)^3 + (z - x)^3$ using suitable algebraic identities.

Ans. (A) To find the value of $48^3 - 30^3 - 18^3$.

Since, $x + y + z = 0$,

then,

$$x^3 + y^3 + z^3 - 3xyz = 0 \text{ or } x^3 + y^3 + z^3 = 3xyz$$



Here, in $48^3 + (-30)^3 + (-18)^3$,

the value of sum,

$$48 + (-30) + (-18) = 0$$

Therefore, $48^3 + (-30)^3 + (-18)^3$

$$= 3 \times 48 \times (-30) \times (-18) = 77760$$

(B) Given $(x - y)^3 + (y - z)^3 + (z - x)^3$

Since, $x + y + z = 0$,

then,

$$x^3 + y^3 + z^3 - 3xyz = 0 \text{ or } x^3 + y^3 + z^3 = 3xyz$$

Here, $(x - y) + (y - z) + (z - x) = 0$

Therefore, $(x - y)^3 + (y - z)^3 + (z - x)^3$

$$= 3(x - y)(y - z)(z - x)$$

LONG ANSWER Type Questions (LA)

[4 & 5 marks each]

8. $2x^3 + 4x^2 - 7ax - 5$ and $2x^3 + ax^2 - 6x + 3$ are polynomials which on dividing by $(x + 1)$ and $(x - 1)$ leaves remainders y and z , respectively, if $y - 3z = 16$ then find a .

Ans. Let, $p(x) = 2x^3 + 4x^2 - 7ax - 5$ and $q(x) = 2x^3 + ax^2 - 6x + 3$ be the given polynomials.

Now,

When $p(x)$ is divided by $(x + 1)$, remainder = y

$$y = p(-1)$$

$$y = 2(-1)^3 + 4(-1)^2 - 7a(-1) - 5$$

$$y = -2 + 4 + 7a - 5$$

$$y = -3 + 7a$$

And, when $q(x)$ is divided by $(x - 1)$, remainder = z

$$z = q(1)$$

$$z = 2(1)^3 + a(1)^2 - 6(1) + 3$$

$$z = 2 + a - 6 + 3$$

$$z = a - 1$$

Substituting the values of y and z , we have

$$y - 3z = 16$$

$$-3 + 7a - 3(a - 1) = 16$$

$$-3 + 7a - 3a + 3 = 16$$

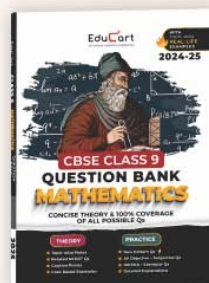
$$4a = 16$$

$$a = 4$$



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